CM5 socket, M.2 connector

File: cm5_socket.kicad_sch

System MCU, Watchdog, EEPROM

File: system_mcu.kicad_sch

AC power, DC power, AC sync

File: ac_dc_power.kicad_sch

Isolated digital IO, Fan controller, Encoder

File: digital_io_fan_encoder.kicad_sch

7-port USB3 Hub

File: usb3_hub.kicad_sch

Rear USB, USB-C

File: usb_conn.kicad_sch

Ethernet, Video outputs, Ref clk

File: video_outputs.kicad_sch

DIB 2.0

File: dib.kicad_sch

Trigger switch matrix, Power modules coupling

File: power_coupling.kicad_sch

USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	FS only	USB #0, HS/SS
Rear bottom	USB A	FS/HS	USB Hub P1 Flex
Front	USB C	USB Hub P2, HS/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, HS/SS	
DIB #2	DIB 2.0	USB Hub P6, HS/SS	
DIB #3	DIB 2.0	USB Hub P5, HS/SS	
DIB #4	DIB 2.0	USB Hub P4, HS/SS	
DIB #5	DIB 2.0	USB Hub P3, HS/SS	

I2C address	Device
0x53	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x3A	PTN5150A USB CC controller
0x70	ADG2188Y 8 x 8 switch array

H1 H2 H3 H4 H5 H6 H7 H8

PWR_FLAG

PE

https://github.com/eez-open/eez-bb3plus

More info at http://www.envox.eu/

Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /

File: BB3plus mainboard.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-03	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 1/10

CM5 socket

Since ID_SD and ID_SC pins are used as GPIO pins, then to prevent the firmware from checking to see if there is a HAT EEPROM available, add force_eeprom_read=0 and disable_poe_fan=1 to the config.txt file.

X1

DF40HC(3.0)-100DS-0.4V(51)

X2

DF40HC(3.0)-100DS-0.4V(51)

CM5_PWR_BTN input:

RasPI CM5: Pull low to force power off or power on from previous software powered off state. Internally pulled up to +5V via 10kΩ

STM32H7 CM5: Reset input, internally pulled up to +3V3 via 10kΩ

CM5_BOOT input:

RasPi CM5: A low on this pin forces booting from an RPI server (e.g. PC or a Raspberry PI). If not used leave floating. Internally pulled up via 10kΩ to CM5_3.3V

STM32H7 CM5: BOOT0 input, internally pulled down to GND via 10kΩ

Module1

Raspberry PI CM5 module silkscreen

M.2 connector

X3A

SM3ZS067U410AMR1000

X3B

USB2 Mux

IC1

PI3USB102ZLEX

S=Low: M input
S=High: D input

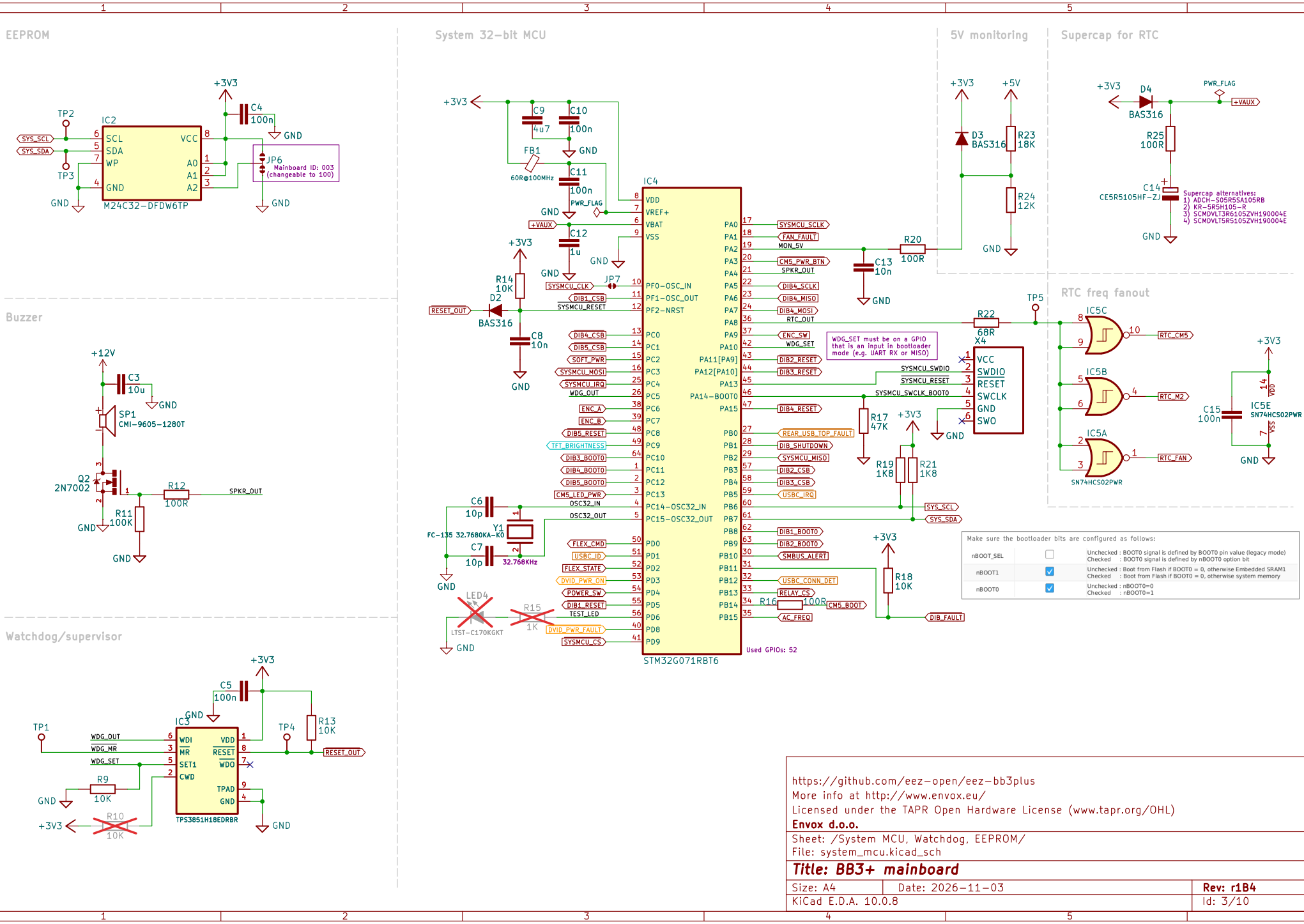
When boot mode is disabled, the USB2 port will be used as a device to access the CM5 via the rpiboot utility. In this state, power must be disabled.

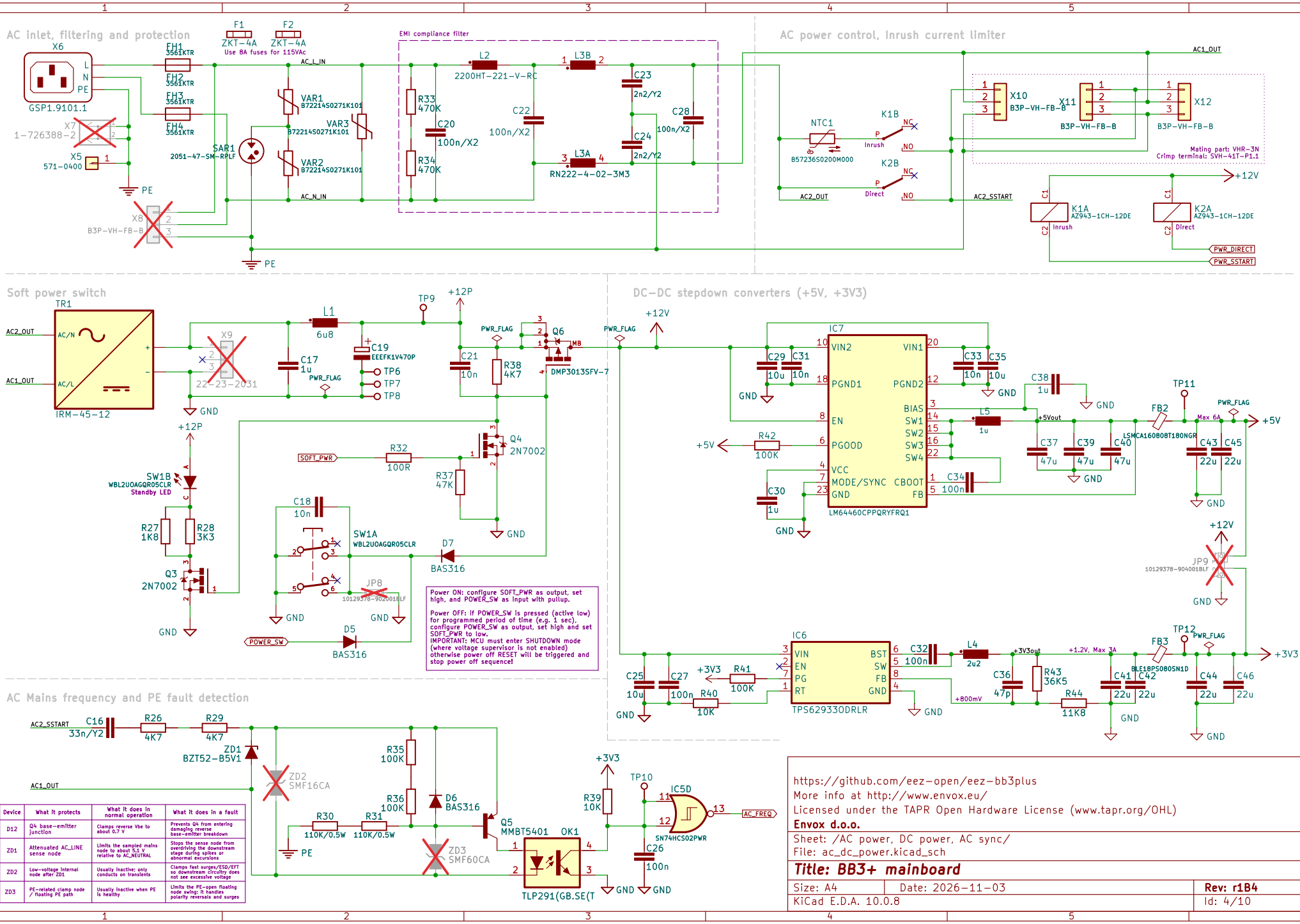
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /CM5 socket, M.2 connector/
File: cm5_socket.kicad_sch

Title: BB3+ mainboard

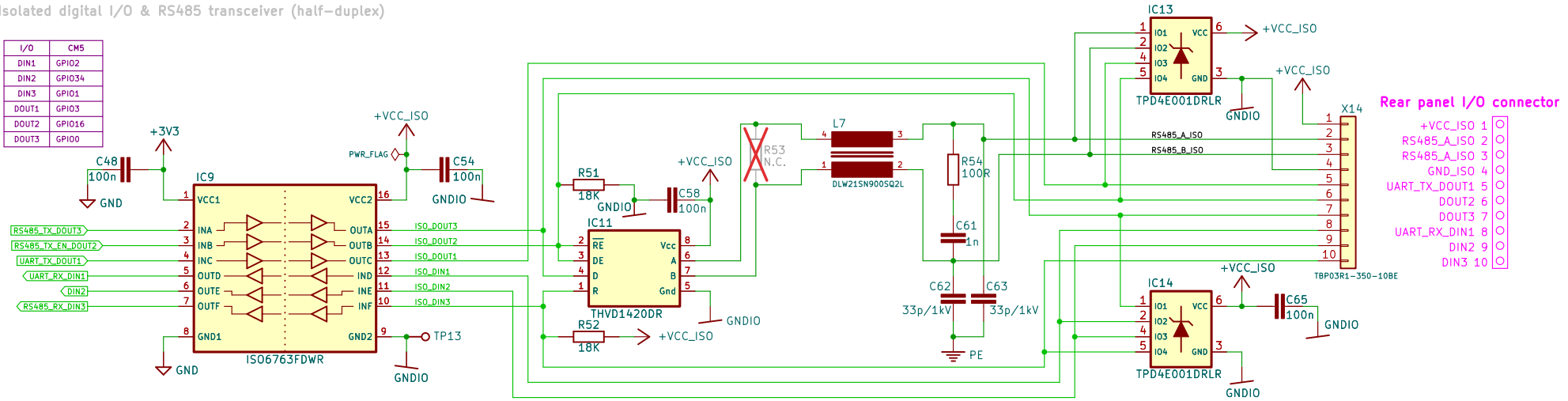
Size: A4	Date: 2026-11-03	Rev: r1B4
KiCad E.D.A. 10.0.8		Id: 2/10


EEPROM
System 32-bit MCU
5V monitoring
Supercap for RTC
Buzzer
RTC freq fanout
Watchdog/supervisor
IC2
M24C32-DFDW6TP
IC4
STM32G071RBT6
IC3
TPS3851H18EDRBR
IC5C
IC5B
IC5A
SN74HCS02PWR
IC5E
SN74HCS02PWR
X4
SP1
CMI-9605-1280T
Q2
2N7002
Y1
FC-135 32.7680KA-K0
32.768KHz
LED4
LTST-C170KGKT
D2
BAS316
D3
BAS316
D4
BAS316
JP6
Mainboard ID: 003
(changeable to 100)
JP7
FB1
60R@100MHz
Supercap alternatives:
1) ADCH-S05R5SA105RB
2) KR-5R5H105-R
3) SCMDVLT3R6105ZVH190004E
4) SCMDVLT5R5105ZVH190004E
WDG_SET must be on a GPIO that is an input in bootloader mode (e.g. UART RX or MISO)
Used GPIOs: 52
Make sure the bootloader bits are configured as follows:
nBOOT_SEL
nBOOT1
nBOOT0
Unchecked : BOOT0 signal is defined by BOOT0 pin value (legacy mode)
Checked : BOOT0 signal is defined by nBOOT0 option bit
Unchecked : Boot from Flash if BOOT0 = 0, otherwise Embedded SRAM1
Checked : Boot from Flash if BOOT0 = 0, otherwise system memory
Unchecked : nBOOT0=0
Checked : nBOOT0=1
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /System MCU, Watchdog, EEPROM/
File: system_mcu.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-03
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 3/10


AC inlet, filtering and protection
F1
ZKT-4A
F2
ZKT-4A
Use 8A fuses for 115VAc
EMI compliance filter
X6
GSP1.9101.1
FH1
3561KTR
FH2
3561KTR
FH3
3561KTR
FH4
3561KTR
AC_L_IN
AC_N_IN
VAR1
B72214S0271K101
VAR3
B72214S0271K101
VAR2
B72214S0271K101
SAR1
2051-47-SM-RPLF
L2
2200HT-221-V-RC
L3B
L3A
RN222-4-02-3M3
R33
470K
R34
470K
C20
100n/X2
C22
100n/X2
C23
2n2/Y2
C24
2n2/Y2
C28
100n/X2
X7
1-726388-2
X5
571-0400
X8
B3P-VH-FB-B
PE
AC power control, Inrush current limiter
NTC1
B57236S0200M000
K1B
K2B
Inrush
Direct
AC2_OUT
AC2_SSTART
AC1_OUT
X10
B3P-VH-FB-B
X11
B3P-VH-FB-B
X12
B3P-VH-FB-B
Mating part: VHR-3N
Crimp terminal: SVH-41T-P1.1
K1A
AZ943-1CH-12DE
K2A
AZ943-1CH-12DE
+12V
PWR_DIRECT
PWR_SSTART
Soft power switch
TR1
IRM-45-12
AC2_OUT
AC1_OUT
X9
22-23-2031
L1
6u8
C17
1u
C19
EEEFK1V470P
TP6
TP7
TP8
TP9
+12P
C21
10n
R38
4K7
PWR_FLAG
Q6
DMP3013SFV-7
GND
Q4
2N7002
SOFT_PWR
R32
100R
R37
47K
SW1B
WBL2UOAGQR05CLR
Standby LED
R27
1K8
R28
3K3
Q3
2N7002
C18
10n
SW1A
WBL2UOAGQR05CLR
JP8
10129378-902001BLF
D7
BAS316
D5
BAS316
POWER_SW
Power ON: configure SOFT_PWR as output, set high, and POWER_SW as input with pullup.
Power OFF: if POWER_SW is pressed (active low) for programmed period of time (e.g. 1 sec), configure POWER_SW as output, set high and set SOFT_PWR to low.
IMPORTANT: MCU must enter SHUTDOWN mode (where voltage supervisor is not enabled) otherwise power off RESET will be triggered and stop power off sequence!
DC-DC stepdown converters (+5V, +3V3)
IC7
LM64460CPPQRYFRQ1
VIN2
VIN1
PGND1
PGND2
BIAS
SW1
SW2
SW3
SW4
EN
PGOOD
VCC
MODE/SYNC
CBOOT
GND
FB
C29
10u
C31
10n
C33
10n
C35
10u
C38
1u
C30
1u
C34
100n
L5
1u
+5Vout
R42
100K
FB2
LSMCA160808T180NGR
Max 6A
TP11
C37
47u
C39
47u
C40
47u
C43
22u
C45
22u
+5V
JP9
10129378-904001BLF
IC6
TPS629330DRLR
VIN
EN
PG
RT
BST
SW
FB
GND
C25
10u
C27
100n
R41
100K
R40
10K
C32
100n
L4
2u2
+3V3out
+1.2V, Max 3A
+800mV
C36
47p
R43
36K5
R44
11K8
FB3
BLE18PS080SN1D
TP12
C41
22u
C42
22u
C44
22u
C46
22u
+3V3
AC Mains frequency and PE fault detection
AC2_SSTART
C16
33n/Y2
R26
4K7
R29
4K7
ZD1
BZT52-B5V1
ZD2
SMF16CA
AC1_OUT
R35
100K
R36
100K
R30
110K/0.5W
R31
110K/0.5W
D6
BAS316
ZD3
SMF60CA
Q5
MMBT5401
OK1
TLP291(GB.SE(T
R39
10K
TP10
IC5D
SN74HCS02PWR
AC_FREQ
C26
100n
Device | What it protects | What it does in normal operation | What it does in a fault
D12 | Q4 base-emitter junction | Clamps reverse Vbe to about 0.7 V | Prevents Q4 from entering damaging reverse base-emitter breakdown
ZD1 | Attenuated AC_LINE sense node | Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL | Stops the sense node from overdriving the downstream stage during spikes or abnormal excursions
ZD2 | Low-voltage internal node after ZD1 | Usually inactive; only conducts on transients | Clamps fast surges/ESD/EFT so downstream circuitry does not see excessive voltage
ZD3 | PE-related clamp node / floating PE path | Usually inactive when PE is healthy | Limits the PE-open floating node swing; it handles polarity reversals and surges
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /AC power, DC power, AC sync/
File: ac_dc_power.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-03
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 4/10

Isolated digital I/O & RS485 transceiver (half-duplex)

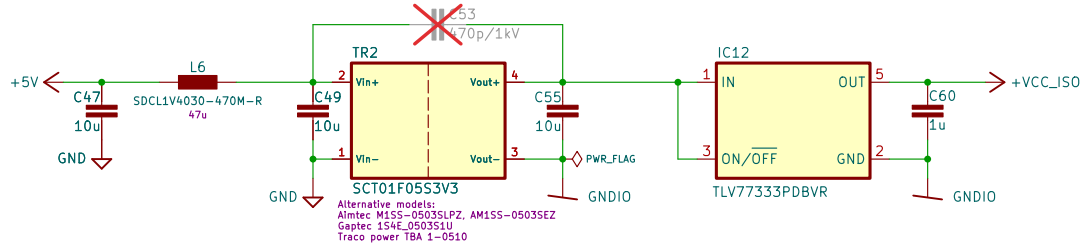
I/O	CM5
DIN1	GPIO2
DIN2	GPIO34
DIN3	GPIO1
DOUT1	GPIO3
DOUT2	GPIO16
DOUT3	GPIO0



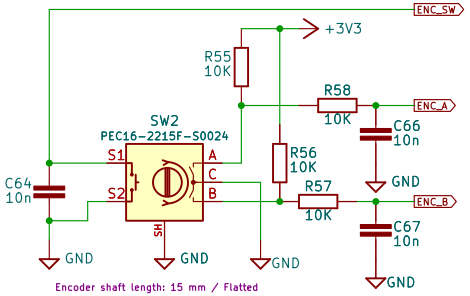
Rear panel I/O connector

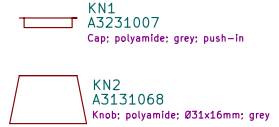
- +VCC_ISO 1
- RS485_A_ISO 2
- RS485_A_ISO 3
- GND_ISO 4
- UART_TX_DOUT1 5
- DOUT2 6
- DOUT3 7
- UART_RX_DIN1 8
- DIN2 9
- DIN3 10

Isolated regulated power supply (+5V)

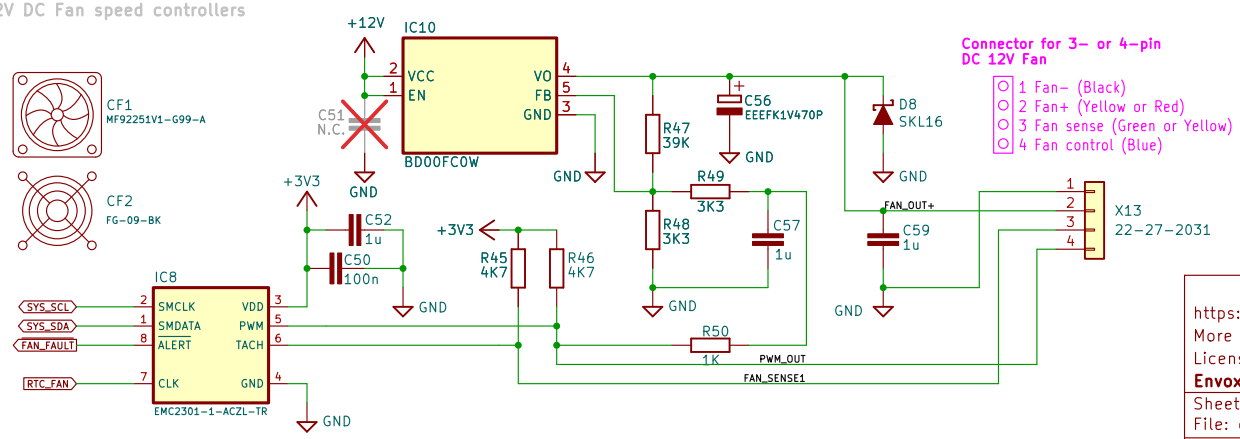


Encoder with switch





12V DC Fan speed controllers



Connector for 3- or 4-pin DC 12V Fan

- 1 Fan- (Black)
- 2 Fan+ (Yellow or Red)
- 3 Fan sense (Green or Yellow)
- 4 Fan control (Blue)

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

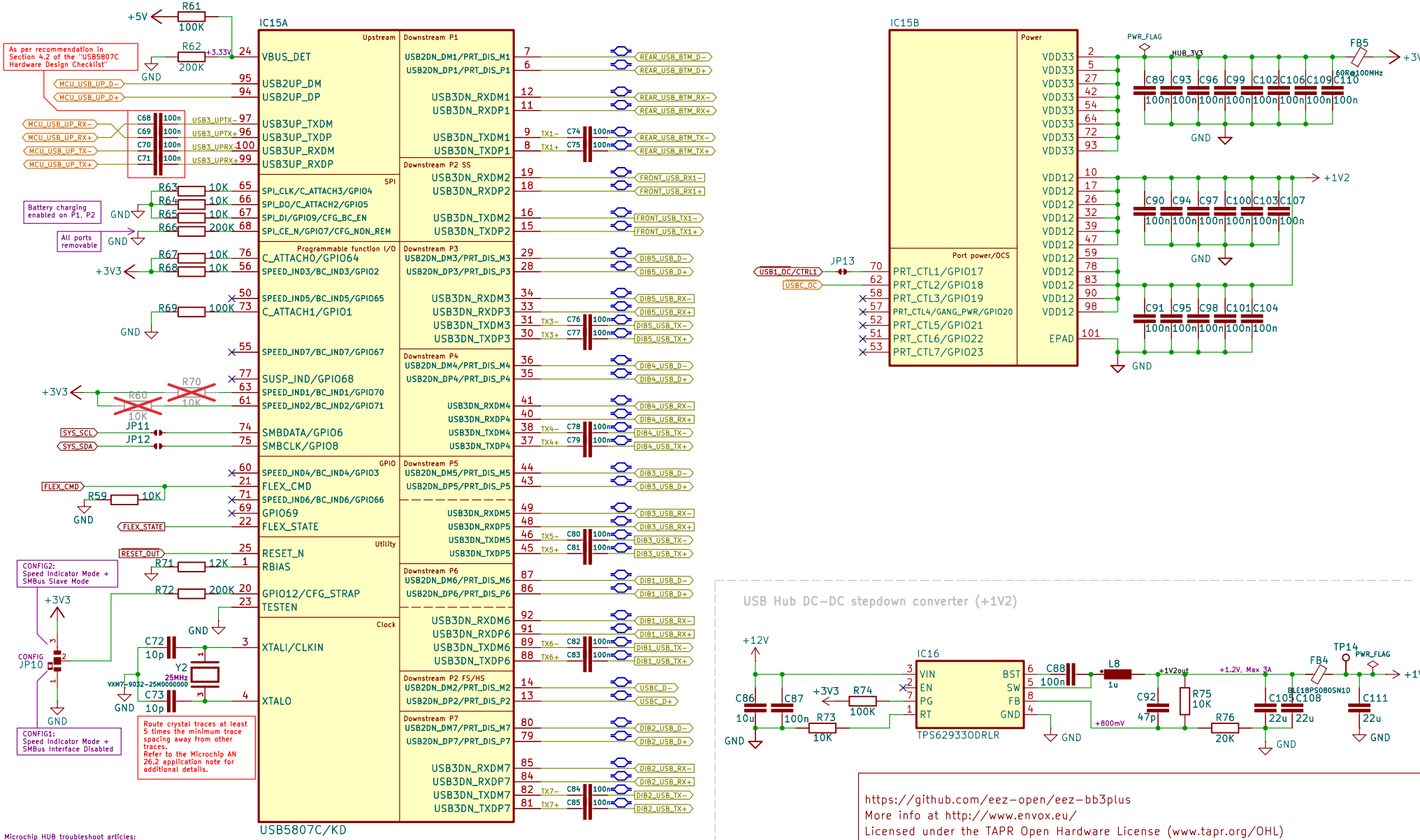
Sheet: /Isolated digital IO, Fan controller, Encoder/
File: digital_io_fan_encoder.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-03	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 5/10

7-port USB hub

All downstream ports that are not used should be disabled (use I2C to configure).



Microchip HUB troubleshoot articles:

https://support.microchip.com/s/article/USB3-Hub-Upstream-Port-Connect-Enumeration-Failure
https://support.microchip.com/s/article/USB-High-Speed-Chirp-Explained
https://support.microchip.com/s/article/USB3-Connection-Process-and-Attach-Detection

IMPORTANT:

If I2C/SMBus pull-up resistors are detected by the USB5807C at start-up, the hub will wait indefinitely to be configured by the attached I2C/SMBus master. For early prototyping, it may be necessary to physically remove the pull-up resistors until the I2C/SMBus master is fully operational and able to properly configure the hub at start-up.

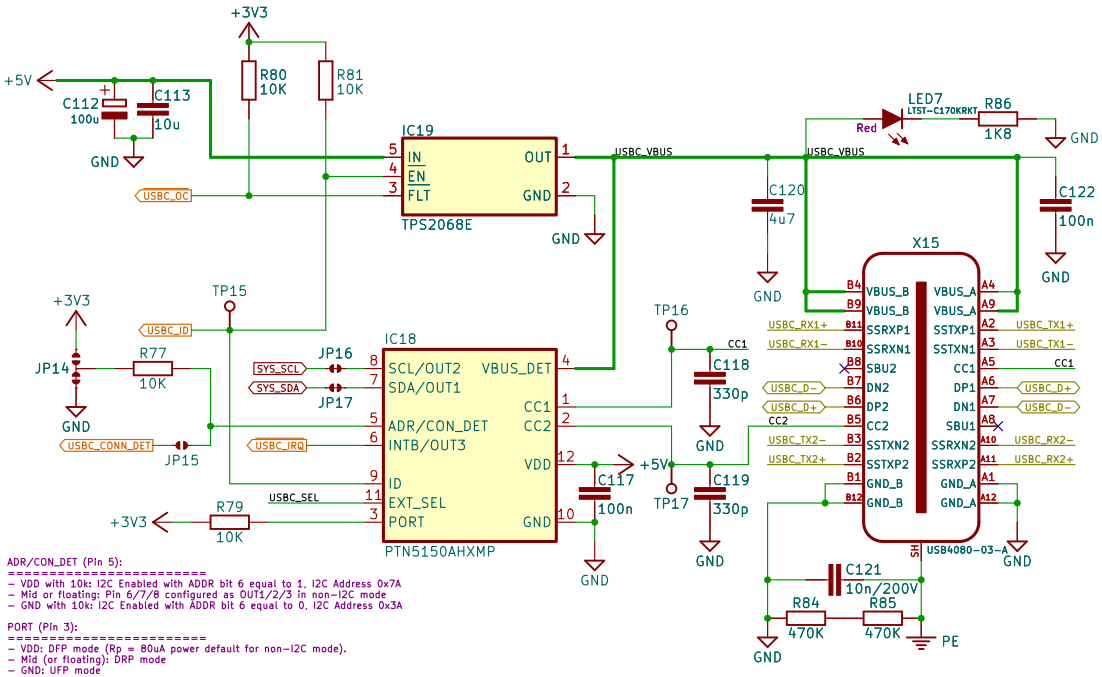
USB Hub DC-DC stepdown converter (+1V2)

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /7-port USB3 Hub/		
File: usb3_hub.kicad_sch		
Title: BB3+ mainboard		
Size: A4	Date: 2026-11-03	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 6/10

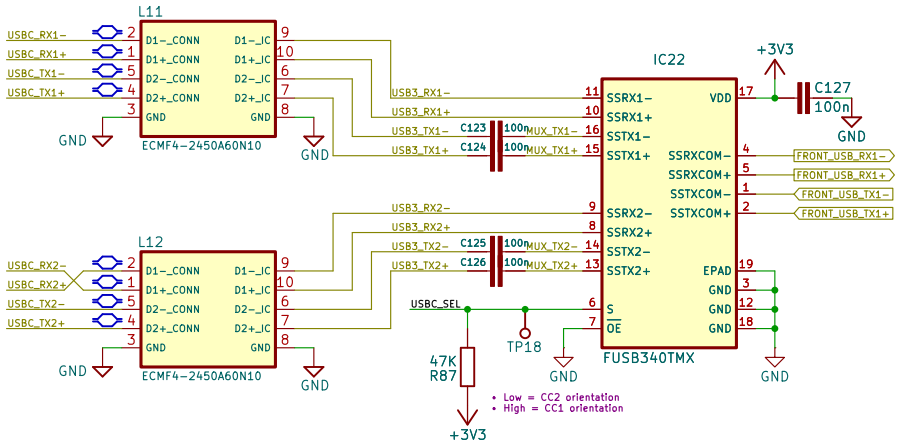
USB-C 3.0 front port (source max 1.5A)



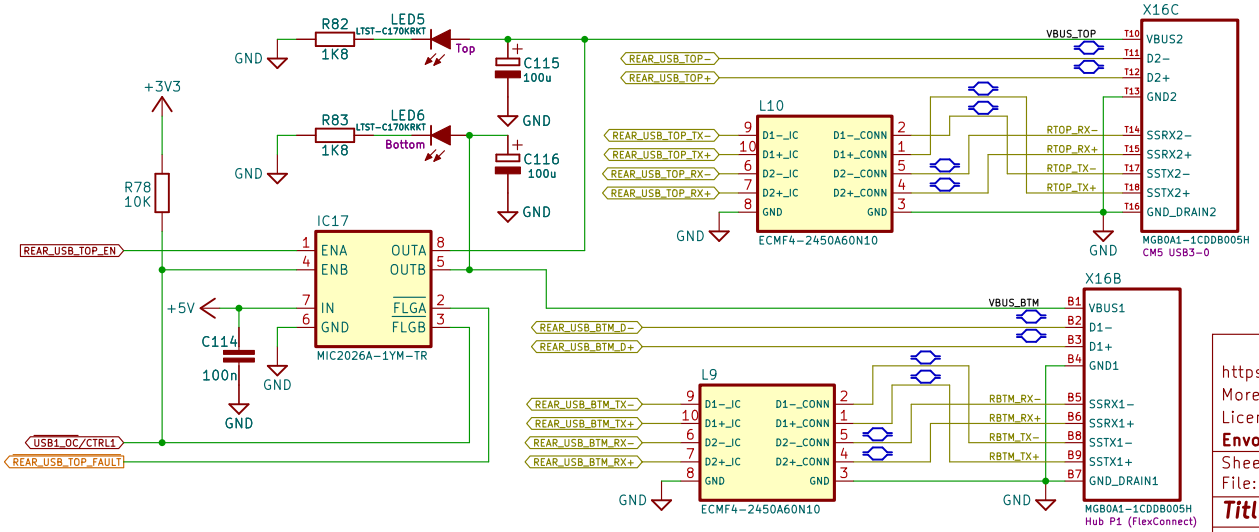
ADR/CON_DET (Pin 5):
- VDD with 10k: I2C Enabled with ADDR bit 6 equal to 1, I2C Address 0x7A
- Mid or floating: Pin 6/7/8 configured as OUT1/2/3 in non-I2C mode
- GND with 10k: I2C Enabled with ADDR bit 6 equal to 0, I2C Address 0x3A

PORT (Pin 3):
- VDD: DFP mode (Rp = 80uA power default for non-I2C mode).
- Mid (or floating): DRP mode
- GND: UFP mode

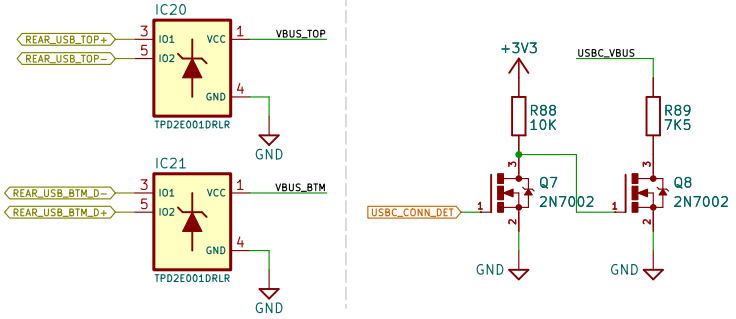
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



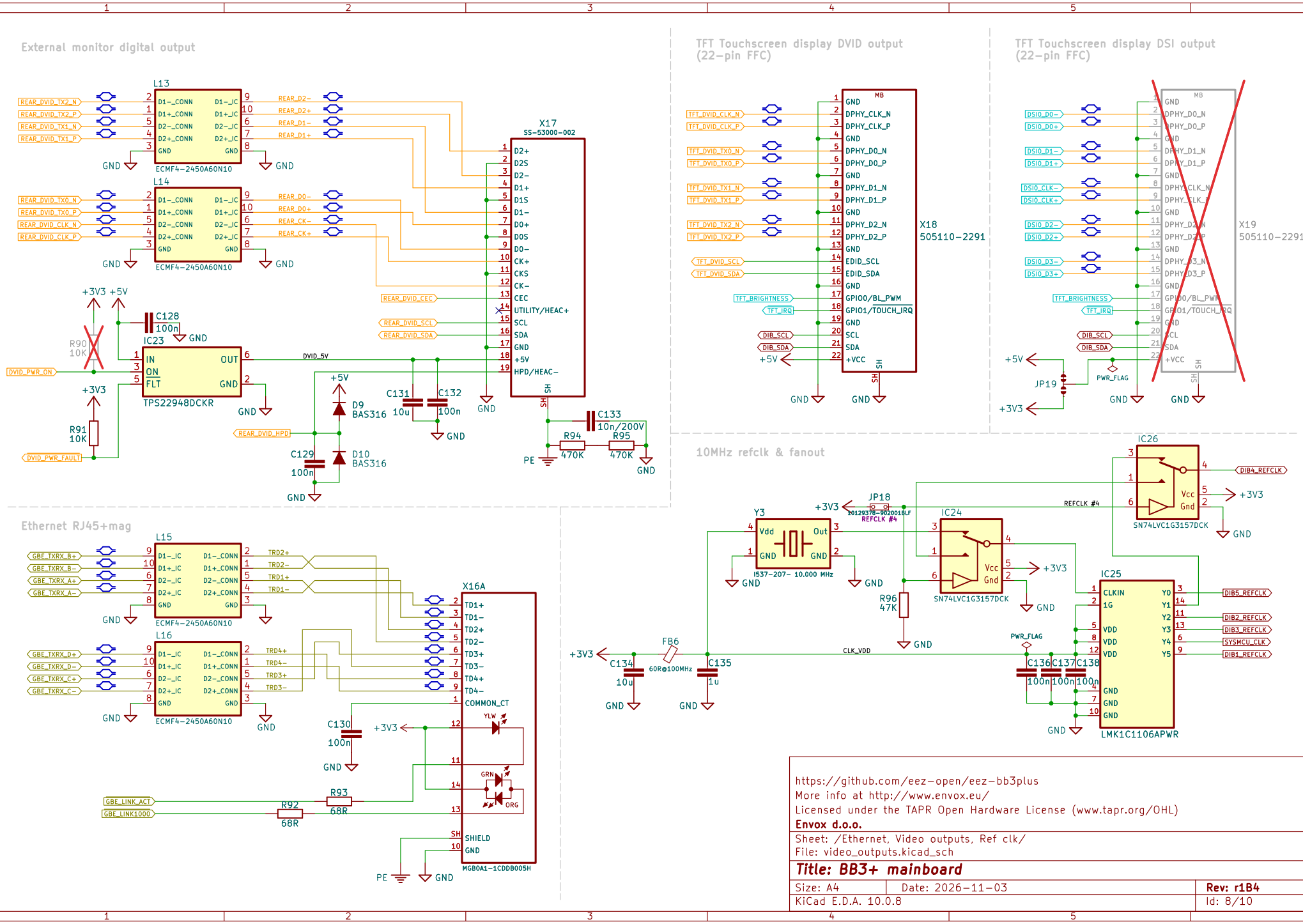
VBus Discharge Circuit

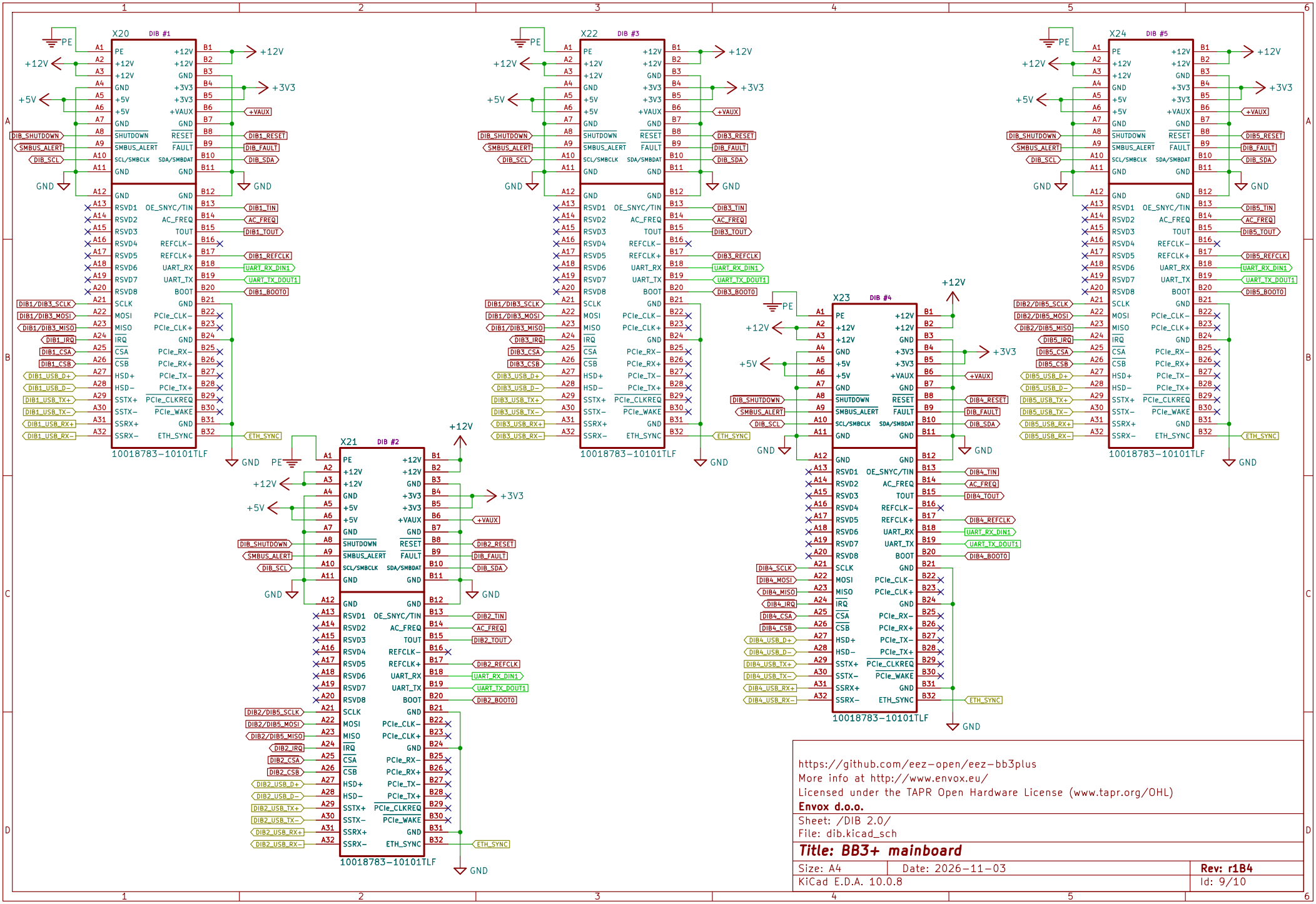


https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /Rear USB, USB-C/		
File: usb_conn.kicad_sch		
Title: BB3+ mainboard		
Size: A4	Date: 2026-11-03	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 7/10


External monitor digital output
L13
ECMF4-2450A60N10
L14
ECMF4-2450A60N10
X17
SS-53000-002
IC23
TPS22948DCKR
C128
100n
R90
10K
R91
10K
C131
10u
C132
100n
C129
100n
D9
BAS316
D10
BAS316
C133
10n/200V
R94
470K
R95
470K
TFT Touchscreen display DVID output
(22-pin FFC)
X18
505110-2291
TFT Touchscreen display DSI output
(22-pin FFC)
X19
505110-2291
JP19
PWR_FLAG
10MHz refclk & fanout
IC26
SN74LVC1G3157DCK
JP18
Y3
I537-207- 10.000 MHz
IC24
SN74LVC1G3157DCK
R96
47K
FB6
60R@100MHz
C134
10u
C135
1u
IC25
LMK1C1106APWR
C136
100n
C137
100n
C138
100n
Ethernet RJ45+mag
L15
ECMF4-2450A60N10
L16
ECMF4-2450A60N10
X16A
MGB0A1-1CDDB005H
C130
100n
R92
68R
R93
68R
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /Ethernet, Video outputs, Ref clk/
File: video_outputs.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-03
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 8/10



https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /DIB 2.0/
File: dib.kicad_sch

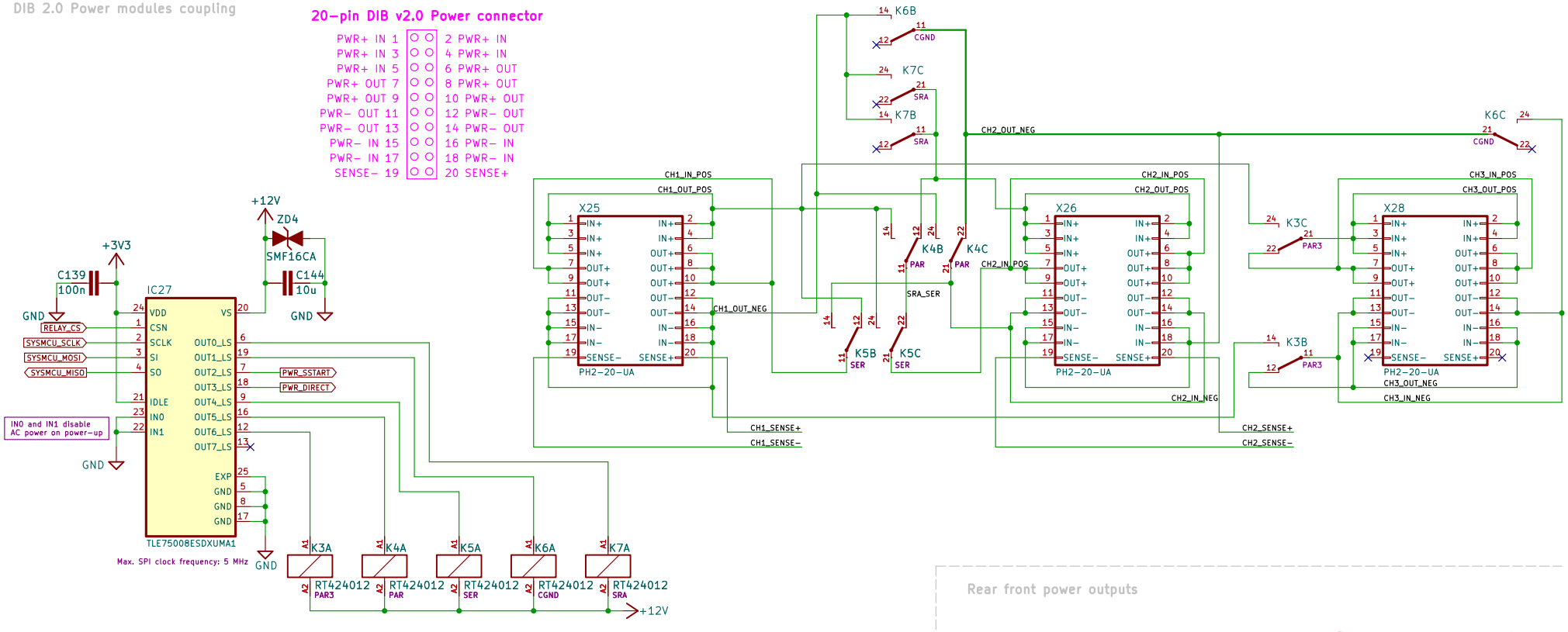
Title: BB3+ mainboard

Size: A4	Date: 2026-11-03	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 9/10

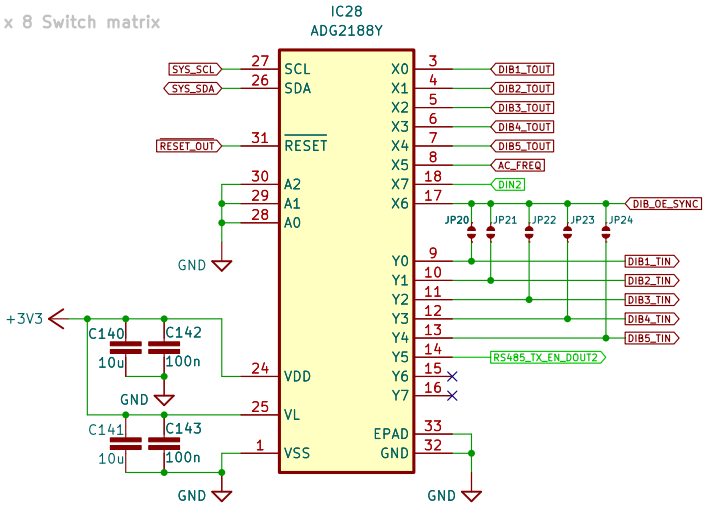
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

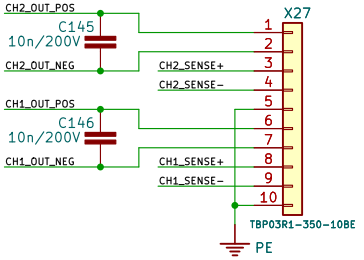
PWR+ IN 1	○ ○	2	PWR+ IN
PWR+ IN 3	○ ○	4	PWR+ IN
PWR+ IN 5	○ ○	6	PWR+ OUT
PWR+ OUT 7	○ ○	8	PWR+ OUT
PWR+ OUT 9	○ ○	10	PWR+ OUT
PWR- OUT 11	○ ○	12	PWR- OUT
PWR- OUT 13	○ ○	14	PWR- OUT
PWR- IN 15	○ ○	16	PWR- IN
PWR- IN 17	○ ○	18	PWR- IN
SENSE- 19	○ ○	20	SENSE+



Trigger 8 x 8 Switch matrix



Rear front power outputs



Rear panel power connector

(Slots #1 and #2 only)

CH2_OUT+ 1	○	2	CH2_OUT-
CH2_SENSE+ 3	○	4	CH2_SENSE-
PE 5	○	6	CH1_OUT+
CH1_OUT- 7	○	8	CH1_SENSE+
CH1_SENSE- 9	○	10	PE

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /Trigger switch matrix, Power modules coupling/
File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-03	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 10/10